

OOPS ASSIGNMENT - 10

- 1)1. Write a Java Swing program to create a Login Form using JTextField, JPasswordField, JButton, and JLabel. When the login button is clicked, validate if the username is "admin" and the password is "password".

```
import java.util.Scanner;

public class LoginConsole {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter username: ");
        String username = scanner.nextLine();

        System.out.print("Enter password: ");
        String password = scanner.nextLine();

        if ("admin".equals(username) && "password".equals(password)) {
            System.out.println("Login successful!");
        } else {
            System.out.println("Invalid username or password.");
        }

        scanner.close();
    }
}
```

```
Enter username: nidhi
Enter password: 2005
Invalid username or password.
```

```
=== Code Execution Successful ===
```

2) Design a simple calculator using Java Swing with buttons for digits (0-9), addition (+), subtraction (-), multiplication (*), and division (/). Implement event handling for button clicks and display the result in a JTextField.

```
import java.util.Scanner;

public class CalculatorConsole {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Simple Calculator");
        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print("Enter operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);

        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();

        double result;

        switch (operator) {
            case '+':
                result = num1 + num2;
                break;

            case '-':
                result = num1 - num2;
                break;

            case '*':
                result = num1 * num2;
                break;

            case '/':
                if (num2 != 0)
                    result = num1 / num2;
                else {
                    System.out.println("Error: Cannot divide by zero.");
                }
            }
        }
```

```
        return;
    }
    break;

default:
    System.out.println("Invalid operator.");
    return;
}

System.out.println("Result: " + result);
}
}
```

Simple Calculator

Enter first number: 56

Enter operator (+, -, *, /): *

Enter second number: 78

Result: 4368.0

=== Code Execution Successful ===

- 3)Write a Java Swing program to implement a To-Do List using JList.
Provide:
- a.A JTextField to enter tasks
 - b.An Add button to add tasks to the list
 - c.A Remove button to delete selected tasks

```
import java.util.ArrayList;
import java.util.Scanner;
```

```
public class ToDoListConsole {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        ArrayList<String> tasks = new ArrayList<>();
        int choice;

        do {
            System.out.println("\n--- To-Do List ---");
            System.out.println("1. View Tasks");
            System.out.println("2. Add Task");
            System.out.println("3. Remove Task");
            System.out.println("4. Exit");
            System.out.print("Enter your choice (1-4): ");
            choice = scanner.nextInt();
            scanner.nextLine(); // Clear the newline

            switch (choice) {
                case 1:
                    System.out.println("\nYour Tasks:");
                    if (tasks.isEmpty()) {
                        System.out.println("No tasks yet.");
                    } else {
                        for (int i = 0; i < tasks.size(); i++) {
                            System.out.println((i + 1) + ". " + tasks.get(i));
                        }
                    }
                    break;

                case 2:
                    System.out.print("Enter a task to add: ");
                    String newTask = scanner.nextLine();
                    tasks.add(newTask);
```

```

        System.out.println("Task added.");
        break;

    case 3:
        System.out.print("Enter the task number to remove: ");
        int index = scanner.nextInt() - 1;
        if (index >= 0 && index < tasks.size()) {
            tasks.remove(index);
            System.out.println("Task removed.");
        } else {
            System.out.println("Invalid task number.");
        }
        break;

    case 4:
        System.out.println("Exiting To-Do List. Goodbye!");
        break;

    default:
        System.out.println("Invalid choice.");
    }

    } while (choice != 4);

    scanner.close();
}
}

```

```

--- To-Do List ---

```

```

1. View Tasks

```

```

2. Add Task

```

```

3. Remove Task

```

```

4. Exit

```

```

Enter your choice (1-4): 2

```

```

Enter a task to add: complete the assignment

```

```

Task added.

```

4) Write a Java Swing program to create a Registration Form that collects user details such as Name, Email, Password, and Confirm Password. Use a combination of JTextField for the name and email inputs, JPasswordField for the password inputs, and a JButton labeled "Register". When the Register button is clicked, validate that none of the fields are empty, that the Password and Confirm Password fields match, and that the email appears valid (for example, containing an "@" symbol). Display a success message on the form if all validations pass; otherwise, show appropriate error messages next to the respective fields.

```
import java.util.Scanner;
import java.util.regex.Pattern;

public class RegistrationForm {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Input Fields
        String name, email, password, confirmPassword;

        // Get Name
        System.out.print("Enter your Name: ");
        name = scanner.nextLine();

        // Get Email
        System.out.print("Enter your Email: ");
        email = scanner.nextLine();

        // Get Password
        System.out.print("Enter your Password: ");
        password = scanner.nextLine();

        // Get Confirm Password
        System.out.print("Confirm your Password: ");
        confirmPassword = scanner.nextLine();
```

```
// Validate inputs
boolean valid = true;

if (name.isEmpty()) {
    System.out.println("Name cannot be empty.");
    valid = false;
}

if (email.isEmpty()) {
    System.out.println("Email cannot be empty.");
    valid = false;
} else if (!isValidEmail(email)) {
    System.out.println("Invalid email format.");
    valid = false;
}

if (password.isEmpty()) {
    System.out.println("Password cannot be empty.");
    valid = false;
}

if (confirmPassword.isEmpty()) {
    System.out.println("Confirm Password cannot be empty.");
    valid = false;
}

if (!password.equals(confirmPassword)) {
    System.out.println("Passwords do not match.");
    valid = false;
}

// Show result
if (valid) {
    System.out.println("\nRegistration Successful!");
    System.out.println("Name: " + name);
    System.out.println("Email: " + email);
} else {
    System.out.println("\nPlease fix the errors above and try again.");
}

scanner.close();
}
```

```
// Email validation function (checks for the '@' symbol and a dot after '@')
private static boolean isValidEmail(String email) {
    String emailPattern = "^[a-zA-Z0-9_+&*-]+(?:\\.[a-zA-Z0-9_+&*-]
+)*@(?:[a-zA-Z0-9-]+\\.)+[a-zA-Z]{2,7}$";
    return Pattern.matches(emailPattern, email);
}
}
```

```
Enter your Name: nidhi
Enter your Email: nidhikesharwani06@gmail.com
Enter your Password: @nidhi2005
Confirm your Password: @nidhi2005

Registration Successful!
Name: nidhi
Email: nidhikesharwani06@gmail.com

=== Code Execution Successful ===
```

5) Write a Java Swing program to implement a simple Stopwatch. The GUI should include a JLabel to display the elapsed time (in seconds) and three JButtons labeled "Start", "Stop", and "Reset". Use a Swing Timer to update the JLabel every second while the stopwatch is running. The "Start" button should begin timing, the "Stop" button should pause the timer, and the "Reset" button should set the elapsed time back to zero. Ensure that the GUI updates correctly based on the user's interactions.

```
import java.util.Scanner;
```

```
public class StopwatchConsole {  
    private static long startTime;  
    private static boolean running = false;  
    private static long elapsedTime = 0;  
  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
  
        while (true) {  
            System.out.println("\n--- Stopwatch ---");  
            System.out.println("1. Start");  
            System.out.println("2. Stop");  
            System.out.println("3. Reset");  
            System.out.println("4. Exit");  
            System.out.print("Enter your choice: ");  
  
            int choice = scanner.nextInt();  
  
            switch (choice) {  
                case 1:  
                    start();  
                    break;  
                case 2:
```

```

        stop();
        break;
    case 3:
        reset();
        break;
    case 4:
        System.out.println("Exiting stopwatch.");
        scanner.close();
        return;
    default:
        System.out.println("Invalid choice. Please try again.");
}

if (running) {
    long currentTime = System.currentTimeMillis();
    elapsedTime = currentTime - startTime;
    System.out.println("Elapsed time: " + elapsedTime / 1000 + "
seconds");
} else {
    System.out.println("Elapsed time: " + elapsedTime / 1000 + "
seconds");
}
}

public static void start() {
    if (!running) {
        startTime = System.currentTimeMillis() - elapsedTime;
        running = true;
        System.out.println("Stopwatch started...");
    } else {
        System.out.println("Stopwatch is already running.");
    }
}

public static void stop() {
    if (running) {
        elapsedTime = System.currentTimeMillis() - startTime;
        running = false;
        System.out.println("Stopwatch stopped.");
    }
}

```

```

    } else {
        System.out.println("Stopwatch is not running.");
    }
}

public static void reset() {
    elapsedTime = 0;
    if (running) {
        startTime = System.currentTimeMillis();
    }
    System.out.println("Stopwatch reset.");
}
}

```

```

--- Stopwatch ---
1. Start
2. Stop
3. Reset
4. Exit
Enter your choice: 1
Stopwatch started...
Elapsed time: 0 seconds

--- Stopwatch ---
1. Start
2. Stop
3. Reset
4. Exit
Enter your choice: 2
Stopwatch stopped.
Elapsed time: 9 seconds

```

